

Signal Processing Lab Work Assignment



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Introduction

In this laboratory session, our goal is to analyse human movement using IMU (accelerometer and gyroscope) and EMG (muscle electrical activity) sensors. The accelerometer measures linear acceleration, which includes both the effects of gravity (approximately 9.81 m/s^2) and dynamic movement forces, making it useful for detecting changes in movement intensity and impact events like jumps. The gyroscope measures angular velocity in degrees per second, capturing rotational movements and providing clear periodic patterns for detecting exercise repetitions such as squats. EMG sensors measure the electrical activity generated by muscle fibres during contraction, allowing us to quantify muscle activation timing and intensity during different movement phases.

Preparation Protocols

- Tester wore shorts to allow proper access to thigh and knee areas for sensor placement
- Sensor placement areas on thigh and knee were shaved to remove hair that interferes with stable sensor-skin contact and causes movement artifacts
- Skin was cleaned thoroughly with alcohol to remove oils, sweat, and dirt that reduce signal quality by preventing firm sensor attachment
- These preparation steps ensured optimal skin-sensor contact essential for accurate IMU and EMG measurements during squat analysis

Note: If tester bleeds after shaving, do not place sensors on those areas.

Sensors

Three Shimmer3 IMU sensors were used during the laboratory session, placed on the pelvis, thigh (midpoint), and calf (midpoint). These sensors measure acceleration, which captures both linear movement forces and the constant effect of gravity (approximately 9.81 m/s^2), as well as angular velocity in degrees per second, which detects rotational movements and provides clear periodic patterns for identifying exercise repetitions such as squats.

One Shimmer3 EMG sensor was placed on the vastus lateralis muscle of the thigh, with the upper electrode positioned at approximately two-thirds of the muscle length, the second electrode placed 2 cm below along the muscle fibres, and a ground electrode on the knee. This sensor measures the electrical activity generated by muscle fibres during contraction, enabling quantification of muscle activation timing and intensity across different phases of the squat movements.

Sensor Placement

Sensors were placed on the pelvis, thigh (midpoint), and calf (midpoint) to capture movement differences across body segments during squats. IMU sensors were attached using bandages (strips) secured firmly to ensure they remained stable relative to the bone during intense physical activity, as loose attachment would cause movement artifacts and inaccurate acceleration or angular velocity readings.

EMG sensors were placed on the vastus lateralis muscle of the thigh with electrodes aligned in one straight line along muscle fibre, upper electrode at approximately two-thirds of muscle length and second electrode 2 cm below, while the ground electrode should be placed on top of knee but we was positioned slightly superior to the knee due to bleeding at the exact knee location, using disposable electrodes for optimal skin contact and reliable electrical signal detection despite repeated contractions.

Proper placement is critical because squats involve rapid, repetitive movements that can dislodge poorly secured sensors, and tight bandage attachment prevents relative motion between sensor and skin while still allowing full range of motion.



Figure 1: Sensor Placements

Laboratory Workflow

My group member commanded and monitored the time using "command and monitor," then the tester performed the following sequence with precise timing:

- Jump (0 s): Reference signal to mark workflow beginning
- Rest (30 s): Tester remained completely still
- 5 slow squats (~30-40 s): Deep, controlled movements
- Rest (30 s): Tester remained completely still
- 5 fast squats (~20-30 s): Clearly faster, explosive movements
- Jump (0 s): Reference signal to mark workflow end

Note: Jumps identify the exact laboratory workflow begin and end because continuous recording captures many background actions. Repeating the protocol gives the best signal quality for analysis. After the final fast squats and jump, there should be no time gap, and the tester should perform no additional movements beyond this protocol.

Thigh Movement Analysis

Signal Processing Workflow

The signal processing workflow began by loading the thigh IMU data from the Shimmer3 sensor our data file with sampling frequency $fs_imu = 128$ Hz, establishing the foundation for systematic analysis of squat biomechanics.

Three sequential steps followed:

- Raw IMU visualization to identify movement phases and select optimal signals
- Gyroscope analysis for precise repetition detection and timing quantification,
- Accelerometer analysis to reveal movement intensity patterns.

This structured approach transformed raw sensor measurements into interpretable biomechanical insights, focusing specifically on thigh signals because they provide the clearest representation of squat cycles while other sensors (hip, calf) were excluded to maintain the individual lab's targeted scope on thigh-dominant movement dynamics.

Step 1: Raw IMU Visualization

We first visualized raw IMU signals from the thigh sensor (accelerometer X, Y, Z and gyroscope X, Y, Z) because this step reveals the complete measurement structure—jumps, rest periods, slow squats, and fast squats.

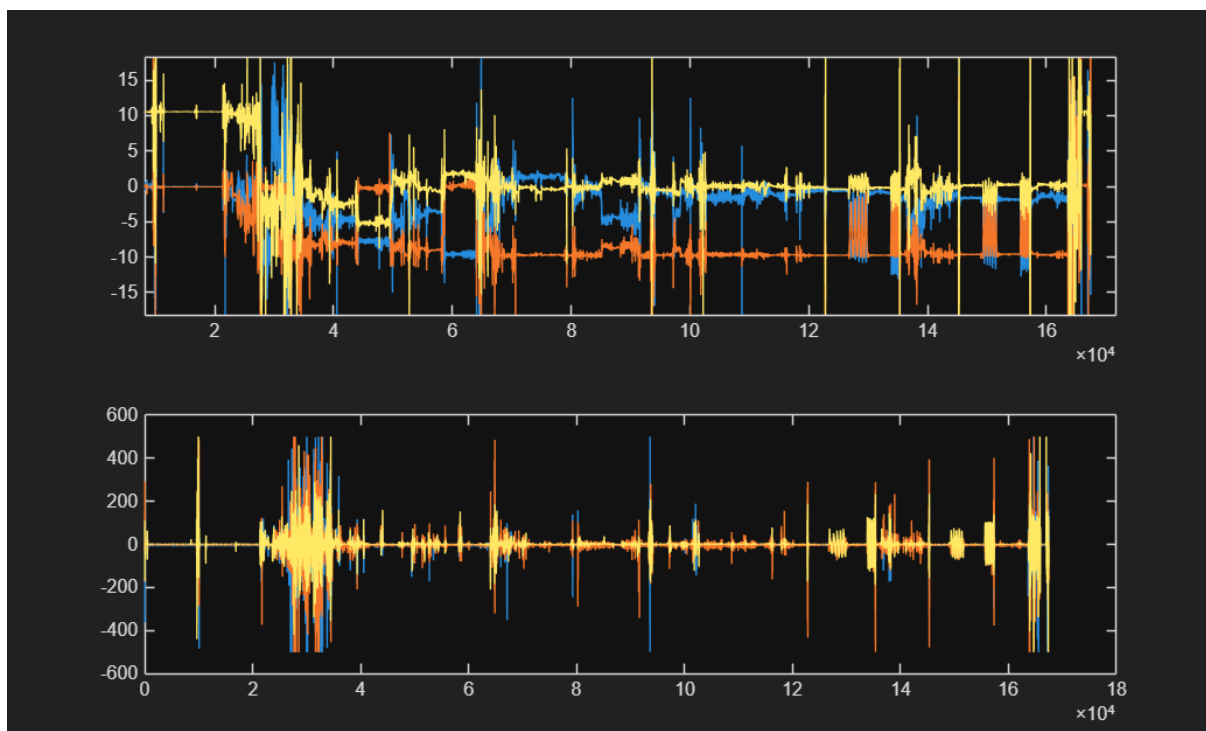


Figure 2:Raw IMU Visualization

Step 2: Gyroscope Analysis

We then processed the thigh gyroscope signal segmented between jumps using peak detection because angular velocity captures rotational thigh movement with sharp, periodic peaks ideal for automatic repetition counting. This step calculated repetitions, peak values, and timing differences between slow/fast squats.

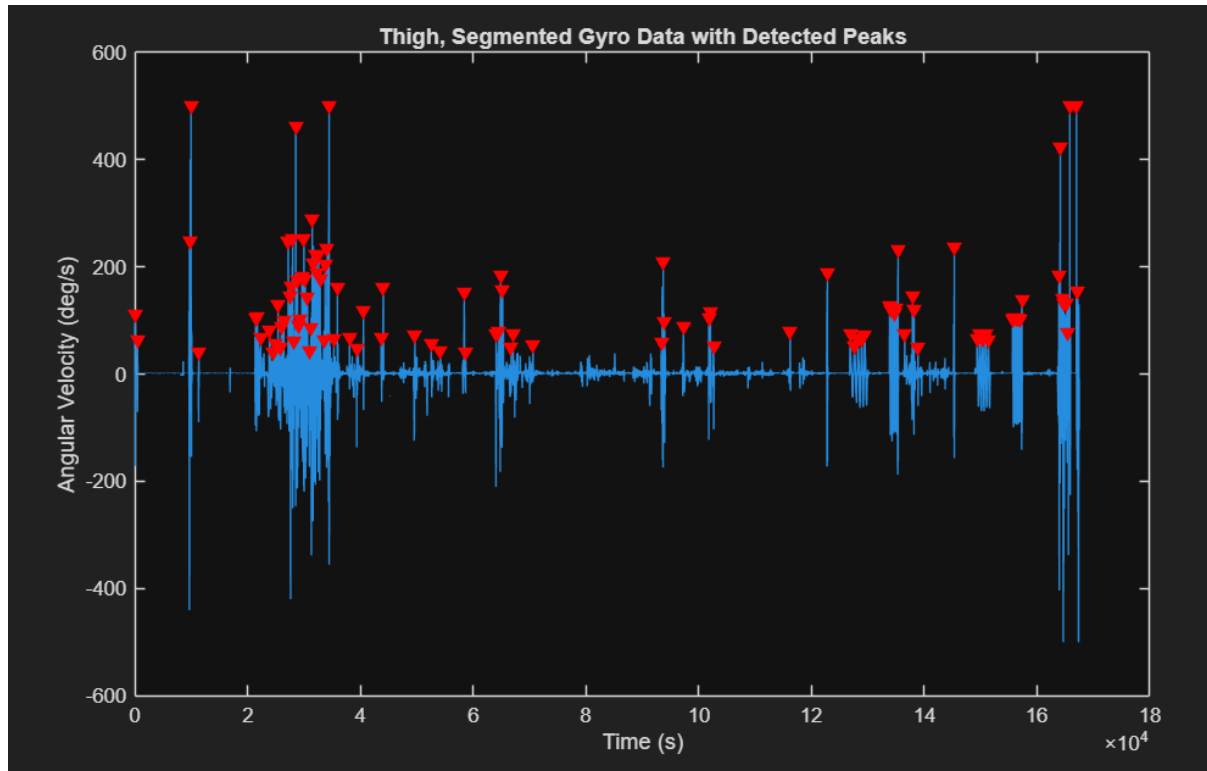


Figure 3: Thigh Segmented Gyro Data with Detected Peaks

We chose gyroscope over accelerometer because it produces distinct cycle peaks while accelerometer magnitude stays gravity-dominated ($\sim 9\text{-}10\text{ m/s}^2$) and lacks clear repetition markers, making gyroscope superior for squat performance quantification.

Step 3: Accelerometer Analysis

Finally, we computed thigh accelerometer magnitude in the same segment to show movement intensity patterns overlaid on the gravitational baseline. This reveals why acceleration remains near $9\text{-}10\text{ m/s}^2$ during rest and how squats create temporary deviations.

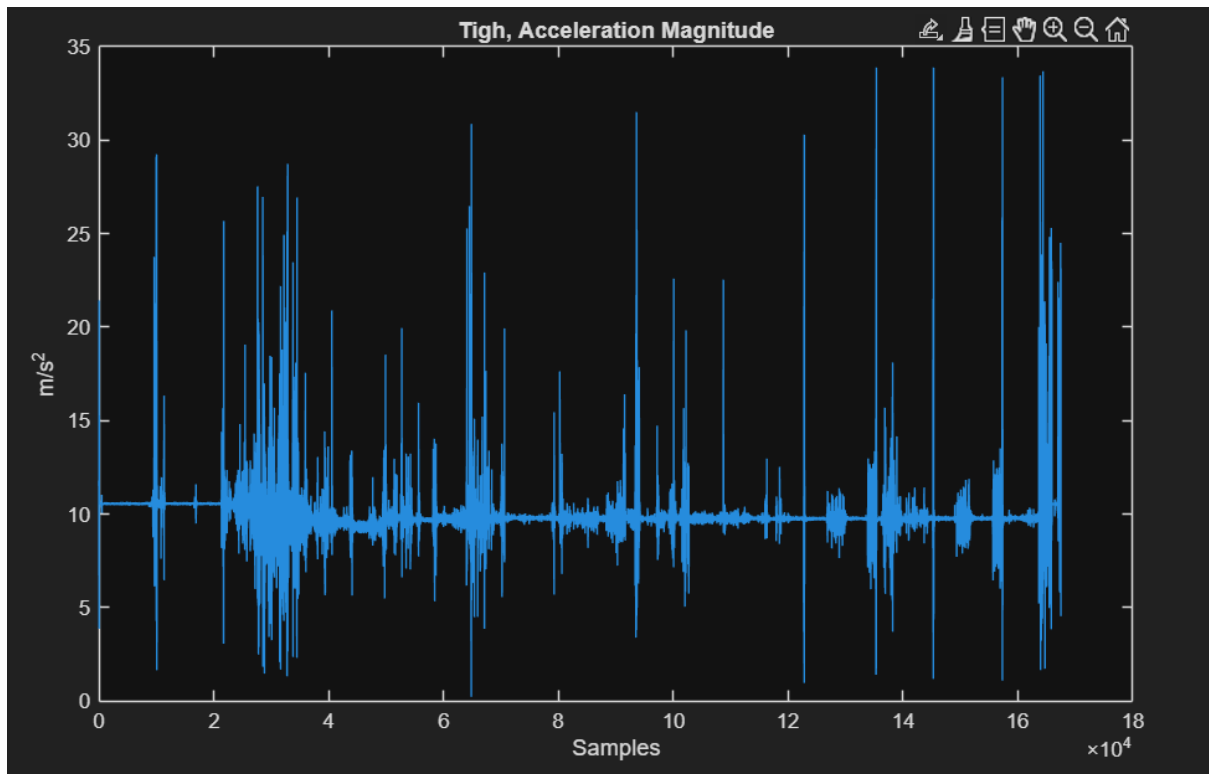


Figure 4: Tigh, Acceleration Magnitude

We analysed accelerometer after gyroscope because it complements rotation data with linear force information, but not as the primary repetition detector due to its smoother profile unsuitable for find peaks. EMG analysis follows separately to link muscle activation to these movement phases.

Result Discussion

Note on Axis Selection: The Z-axis of the thigh gyroscope was chosen for analysis because it showed the clearest periodic squat cycles with highest peak-to-valley contrast among X, Y, and Z axes, corresponding directly to primary knee flexion-extension rotation during squats when the sensor was aligned with the thigh bone.

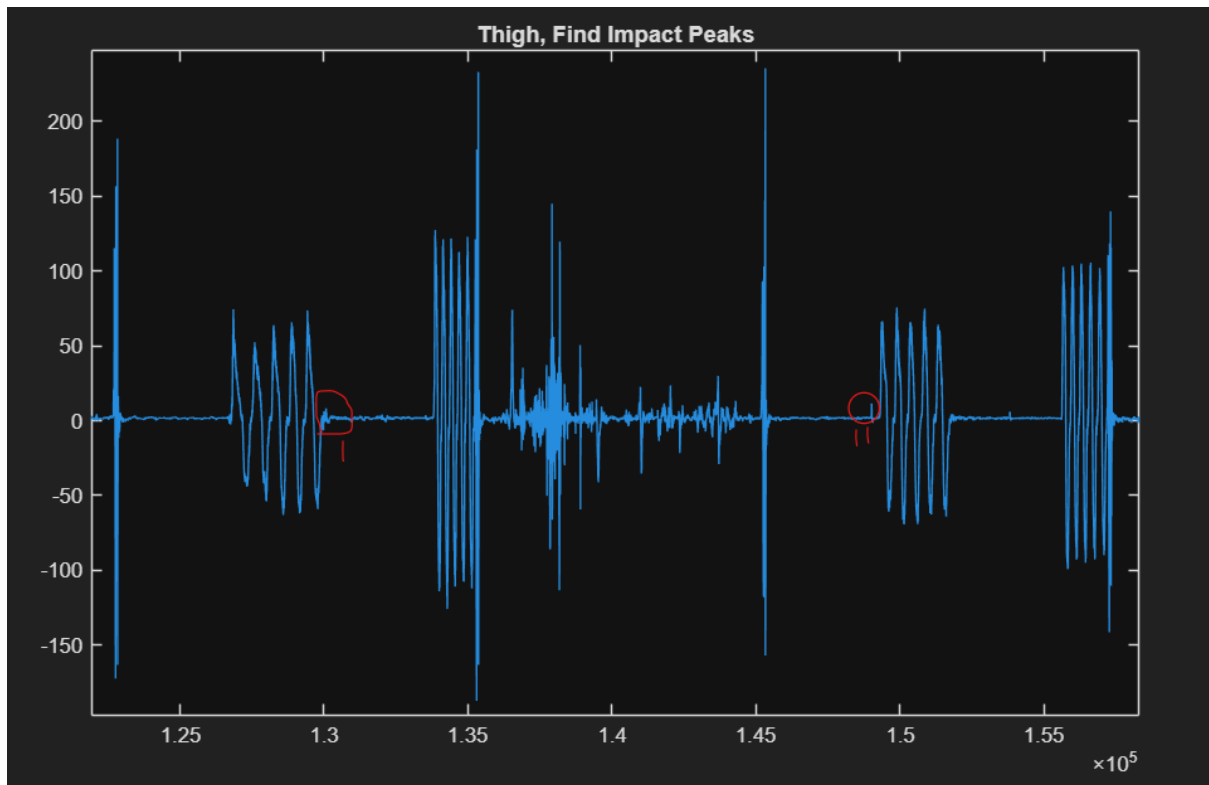


Figure 5: Tigh Find Impact peaks

Figure 5 shows two squat trials separated by jump events used to mark the beginning and end of the measurement period. In the first trial, the signal is noisier because the IMU bandage was loose, which likely caused slight sensor motion relative to the thigh. However, the slow squats are still visible because the tester performed them with longer duration and more controlled timing. The second trial is smoother and more uniform, indicating better sensor stability and more consistent execution, although the squat speed was moderate rather than very fast or very slow. The additional movements marked I and II are not true test actions but small leg adjustments and balance corrections during the test. The last jump is slightly smaller than the others, but it still functions as the end reference point of the protocol.

	Test 1 (1.23 – 1.36)	Test 2 (1.46 – 1.59)
Nature of signal	Bit noisy than other	uniform
Slow squat	Vell illustrated	moderate
High squat	Much fast hard to analyse	moderate

Because of these points I choose Test 2 for analysation.

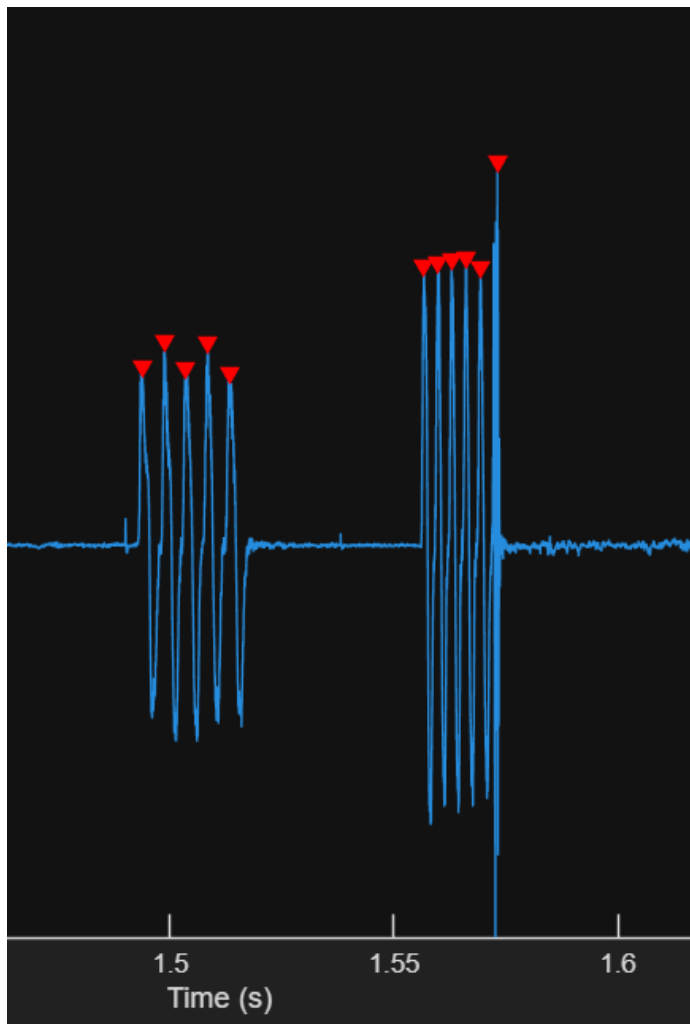


Figure 6: Gyroscope analysis of second test

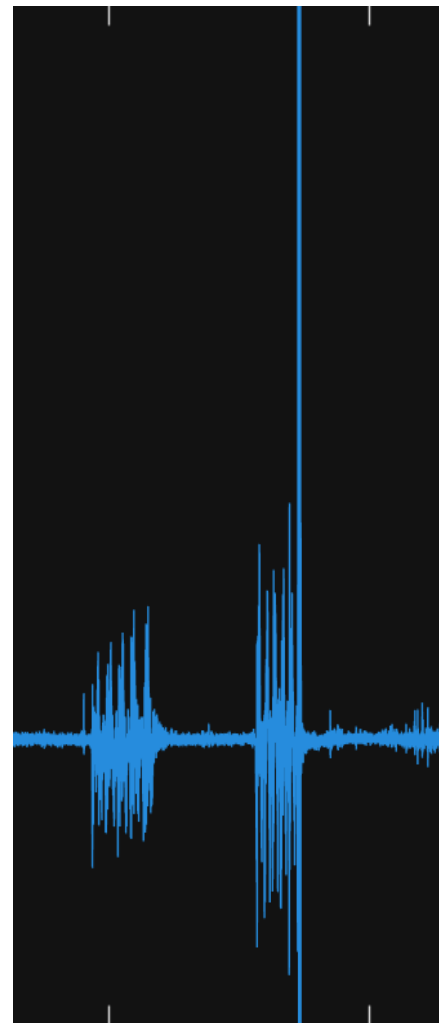


Figure 7: Accelerometer analysis of second test

The gyroscope (figure 6) signal was most useful for repetition detection because it produces sharp, alternating positive and negative peaks that clearly mark each squat cycle, while accelerometer magnitude (figure 7) remains relatively smooth and gravity-dominated without distinct repetition markers.

I selected a single representative waveform from the gyroscope data showing the characteristic pattern where the signal first goes positive during the downward squat phase (thigh flexion/rotation), then negative during the upward return (thigh extension), creating a clear periodic signature that matches the biomechanical squat motion.

Slow squats showed longer peak-to-peak intervals (~2-3 seconds) with lower peak amplitudes, indicating controlled, deliberate thigh rotation, while fast squats had shorter intervals (~1-2 seconds) and higher peaks reflecting quicker angular velocity from explosive movement. These differences demonstrate biomechanically how fast squats require greater rotational speed and power, while slow squats emphasize stability and muscle endurance.

In slow squats 1, 3, and 5, the positive peak values were lower than squats 2 and 4, likely due to slight variations in squat depth, speed, or balance adjustments by the tester during those specific repetitions. Similarly, the 5th fast squat showed reduced amplitude compared to earlier fast squats, possibly indicating fatigue accumulation at the end of the set or a slightly less explosive execution.

Hip & Calf Movement Analysis

As thigh signals from the second test were previously selected for primary analysis due to their smoother quality and better sensor stability, gyroscope results from hip and calf were also processed using the same segmented period between jumps.

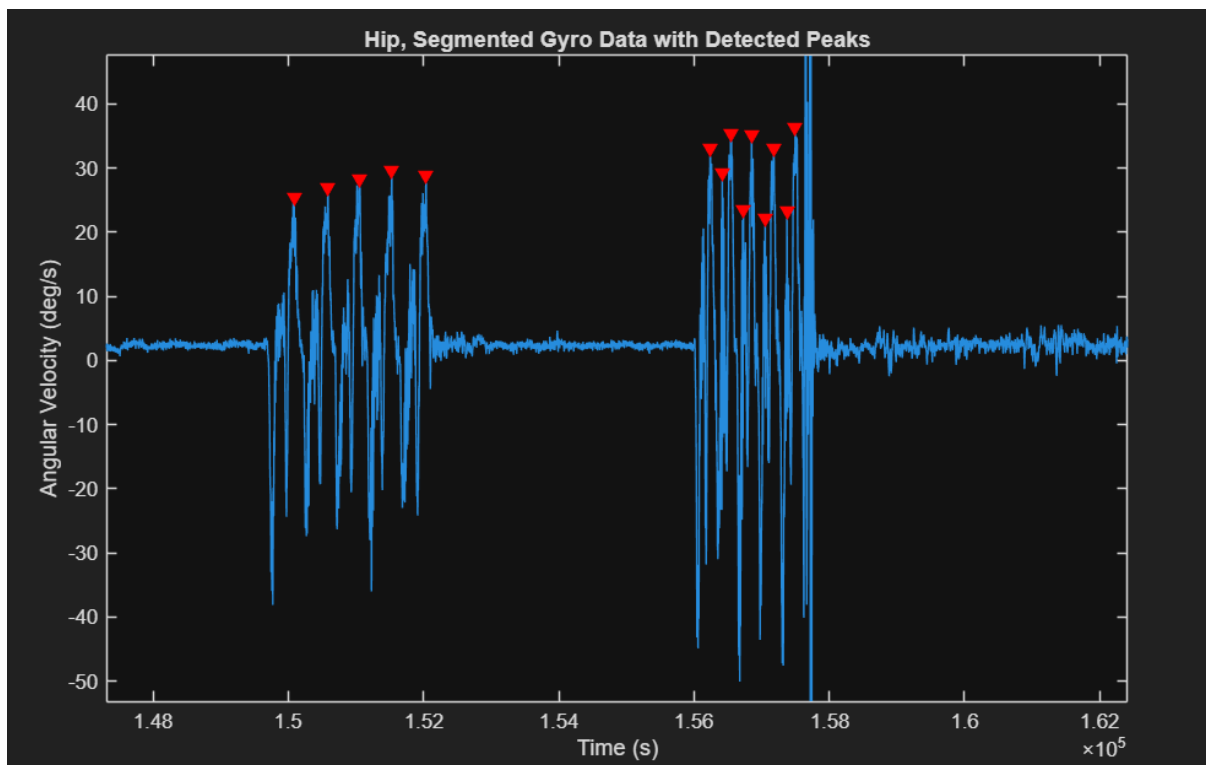


Figure 8: Hip, Segmented Gyro Data with Detected Peaks

Figure 8 shows the hip gyroscope signal with detected peaks during the squat protocol, exhibiting noisier characteristics compared to thigh signals due to the pelvis location experiencing greater soft tissue movement and multi-planar rotation from trunk stabilization efforts. The hip signal appears irregular with smaller amplitude peaks because the pelvis undergoes subtler angular changes during squats compared to the thigh's more pronounced knee-driven flexion-extension, and bandage attachment at the hip may have been less stable due to waist movement and clothing interference. While it generally agrees with the thigh waveform pattern (alternating positive-negative cycles corresponding to squat phases), the hip signal shows more variability and fewer distinct repetitions, confirming thigh gyroscope as superior for precise repetition counting.

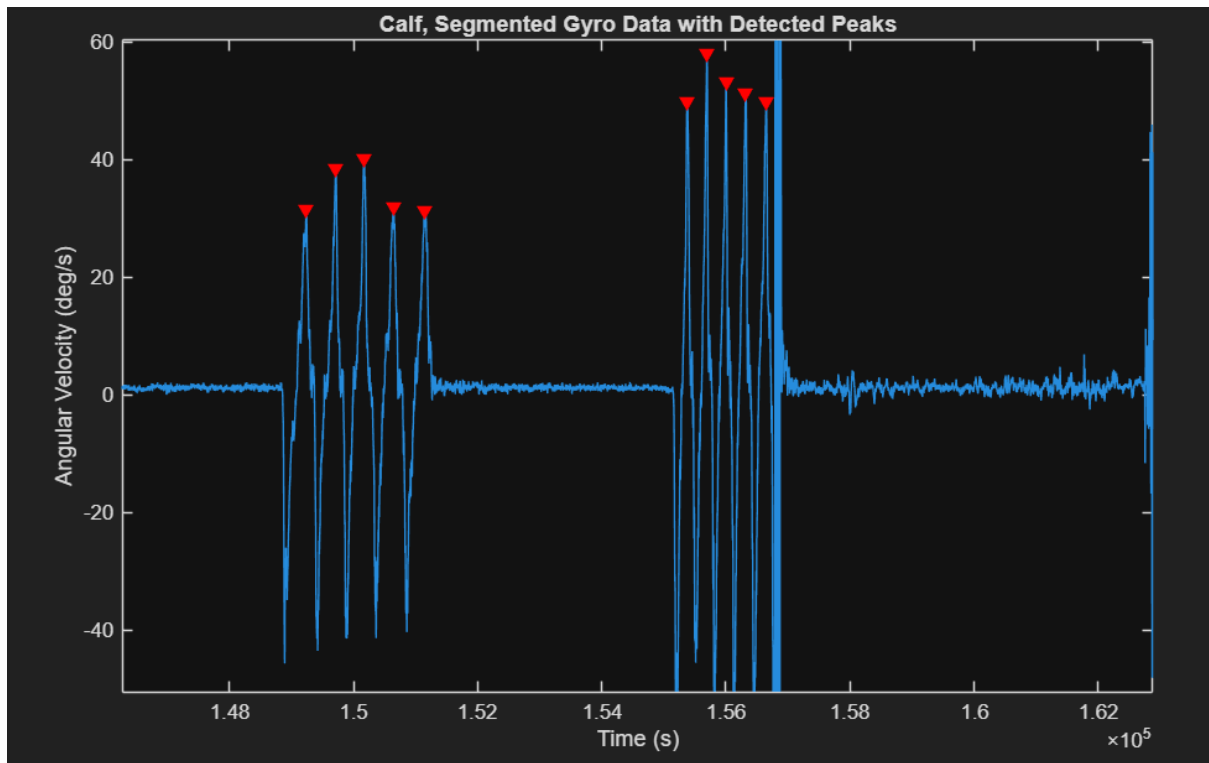


Figure 9: Calf, Segmented Gyro Data with Detected Peaks

Figure 9 displays the calf gyroscope signal, showing cleaner peaks than hip but different timing and amplitude compared to thigh because the calf primarily captures ankle plantarflexion-dorsiflexion rather than knee-dominant thigh rotation. The calf waveform agrees with thigh in showing periodic squat cycles but with shorter peak-to-peak intervals and lower peak heights, reflecting the calf's secondary role in squat biomechanics where gastrocnemius and soleus contribute mainly during push-off phases rather than full flexion-extension. This distal location produces more uniform signals due to better bandage stability around the smaller calf circumference, but lacks the thigh's amplitude clarity for optimal repetition detection, validating the focus on thigh IMU for primary squat analysis.

The thigh gyroscope showed the clearest repetition peaks with highest amplitude and consistent alternating positive-negative cycles corresponding to knee flexion-extension, while hip signals were noisier with smaller peaks due to pelvis soft tissue movement and trunk stabilization demands, and calf signals exhibited cleaner but lower-amplitude peaks reflecting ankle rather than knee motion. This confirms the thigh gyroscope as optimal for squat repetition detection because it captures the primary knee joint rotation most directly, whereas hip reflects secondary stabilization and calf shows distal push-off contributions, both with reduced clarity for automatic peak counting compared to thigh-dominant squat biomechanics.

EMG Signal Processing

Signal Processing Workflow

The EMG signal processing workflow began by loading the thigh EMG data from the Shimmer3 sensor (EMG.mat file) with sampling frequency $fs_emg = 256$ Hz, establishing the foundation for systematic analysis of muscle activation during squat biomechanics. Five sequential steps followed:

- Raw EMG visualization to inspect unprocessed muscle electrical activity
- Band-pass filtering (20-100 Hz) to remove noise while preserving physiological frequencies
- Rectification to convert bipolar signals to positive amplitude values
- RMS calculation to quantify activation intensity
- EMG envelope creation for smooth timing representation.

This structured approach transformed noisy raw muscle signals into interpretable activation patterns, focusing specifically on vastus lateralis EMG because it provides the clearest representation of quadriceps effort during squats while other muscles were excluded to maintain the individual lab's targeted scope on primary knee extensor dynamics.

Raw EMG Visualization

The EMG signal processing began with visualization of the raw EMG signal from the Shimmer3 sensor (EMG.mat file, $fs_emg = 256$ Hz), segmented to match the squat measurement period. This step displayed the unprocessed electrical activity recorded from the vastus lateralis muscle, revealing high-frequency noise, baseline fluctuations, and the characteristic bipolar oscillations produced by muscle fiber action potentials.

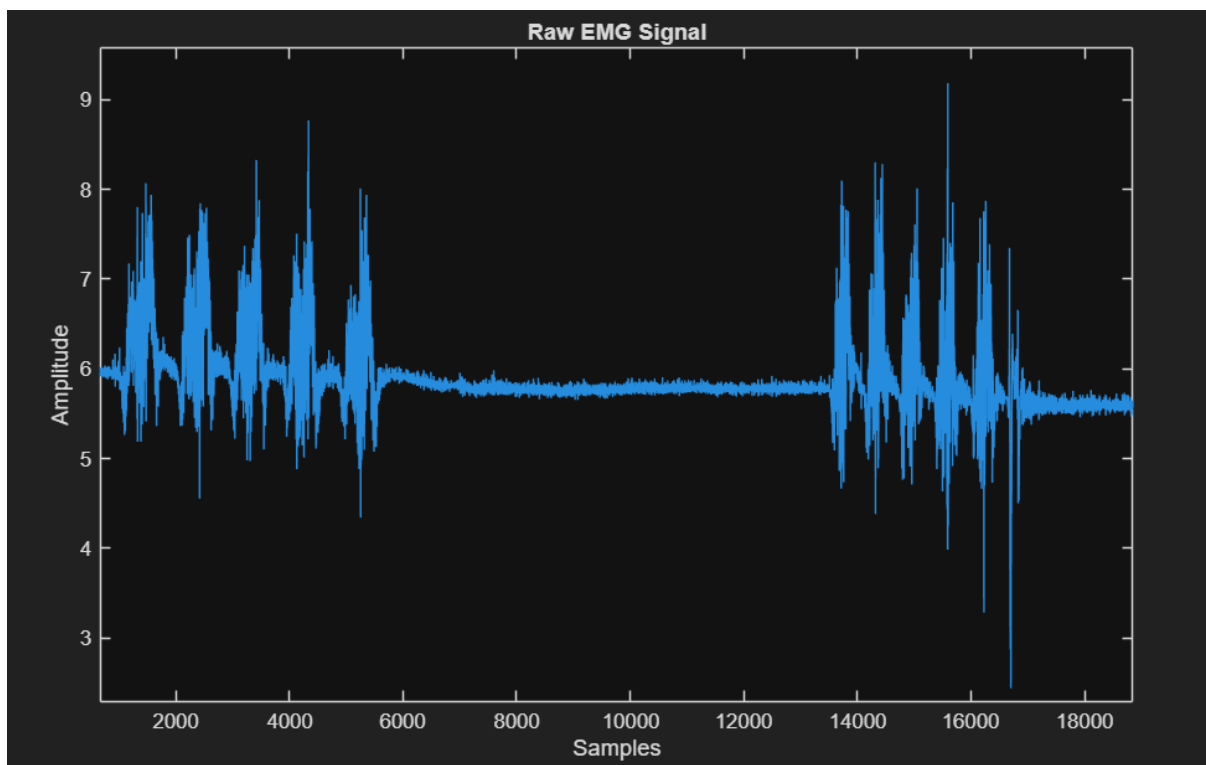


Figure 10: Raw EMG Signal

Band-Pass Filtered EMG

Band-pass filtering (20-100 Hz) was then applied to remove low-frequency movement artifacts and high-frequency power-line noise while preserving the physiological frequency range of muscle activation signals. This step cleaned the raw EMG, making the underlying muscle activation patterns more visible and preparing the signal for subsequent amplitude-based analysis.

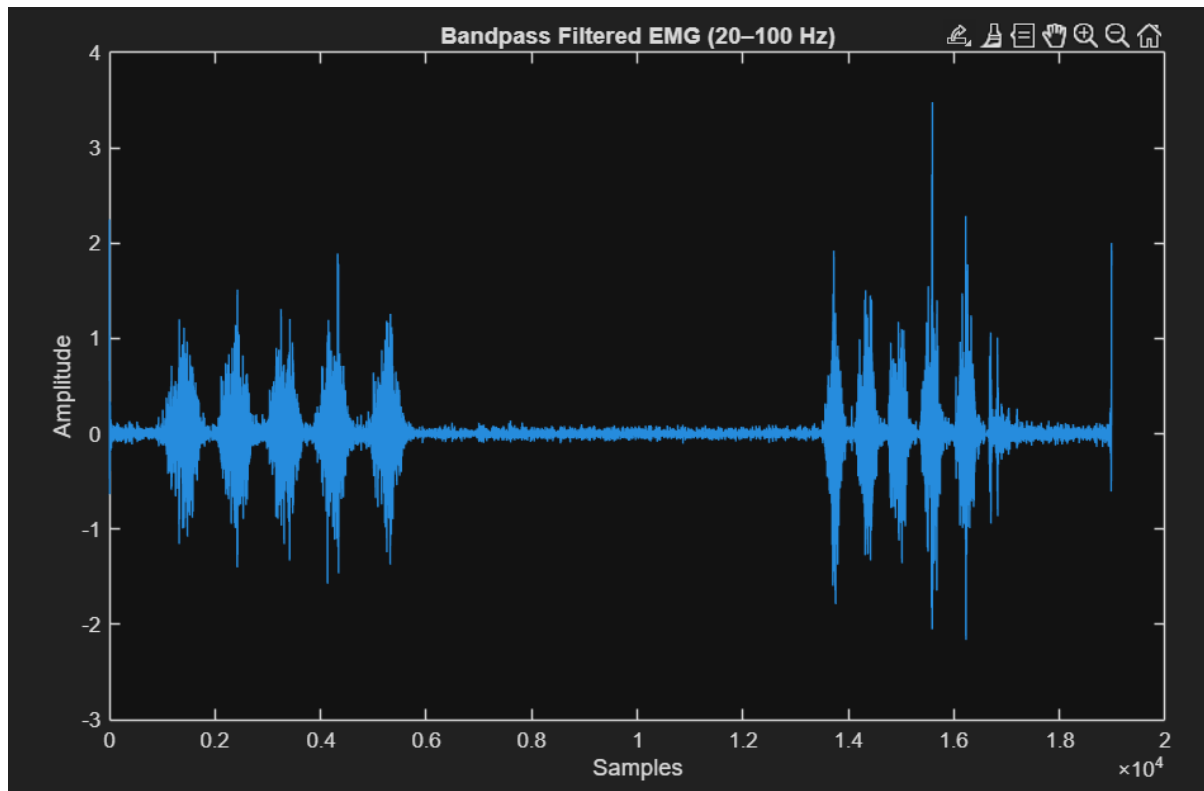


Figure 11: Bandpass Filtered EMG

Rectified EMG

Rectification converted the filtered bipolar signal to positive values only using the absolute value function because muscle activation strength is represented by signal amplitude regardless of polarity. This transformation eliminated negative oscillations, creating a unidirectional signal where peak heights directly correspond to contraction intensity during squat movements.

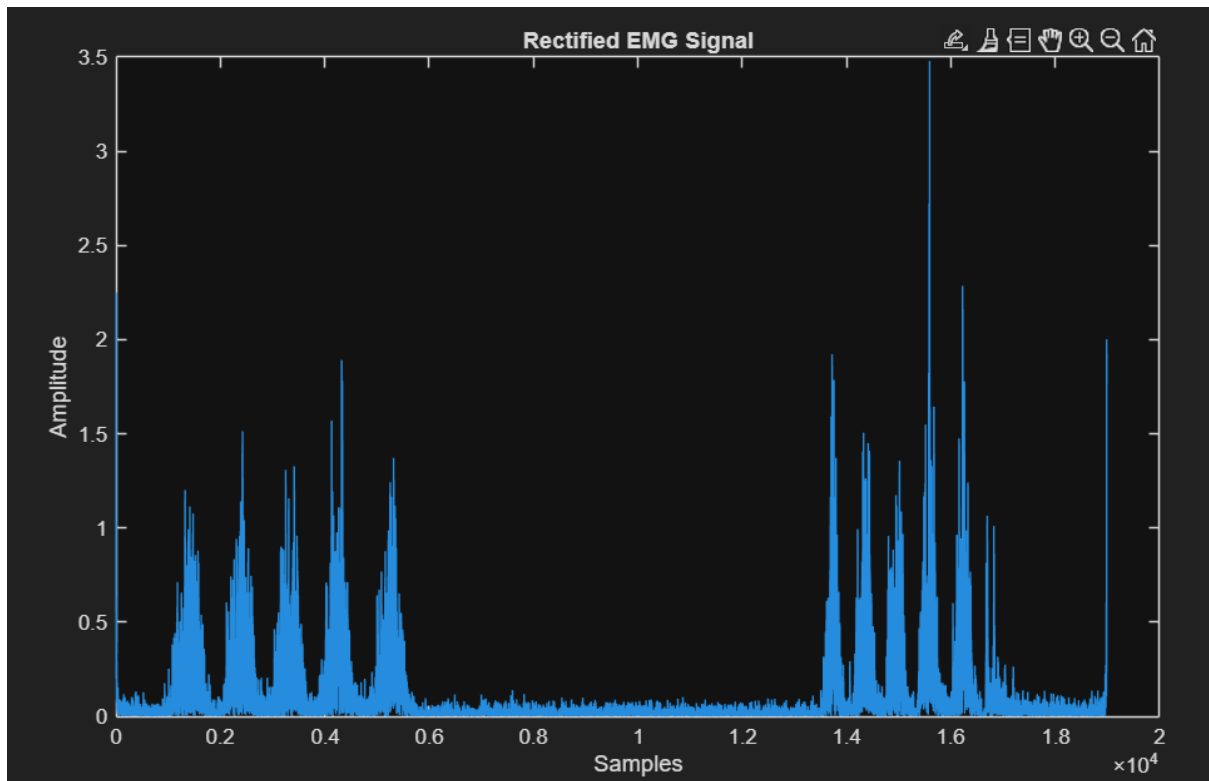


Figure 12: Rectified EMG Signal

RMS Calculation

Root Mean Square (RMS) analysis followed using a 100-sample moving window:

```
rms_signal = sqrt(movmean(emg_rectified.^2, 100))
```

RMS quantifies the signal power as a measure of muscle activation intensity, smoothing minor fluctuations while preserving the overall activation envelope across slow and fast squat phases.

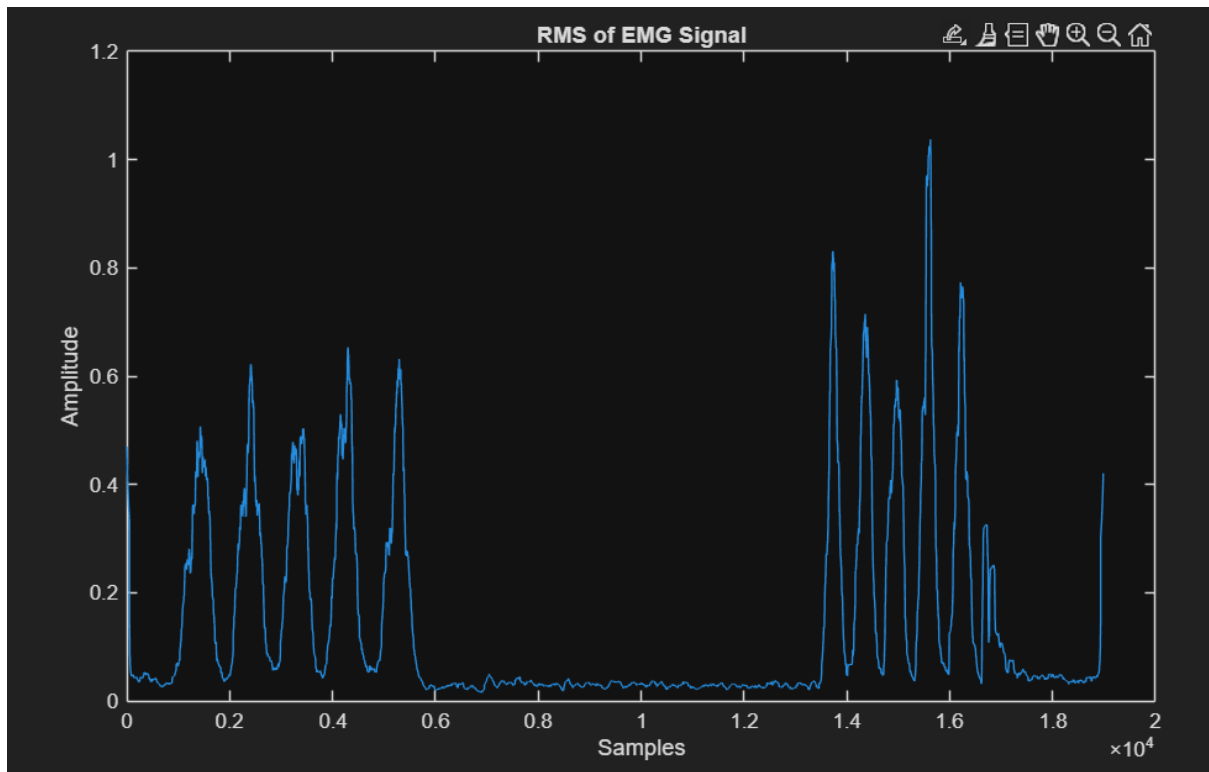


Figure 13: RMS of EMG Signal

EMG Envelope

The final EMG envelope was created using a moving average (e.g., `movmean(emg_rect, 200)`) or MATLAB's `envelope` function to produce a smooth curve representing overall muscle activation timing. This step reveals clear activation peaks synchronized with squat cycles, enabling straightforward comparison of muscle effort between slow (controlled) and fast (explosive) conditions.

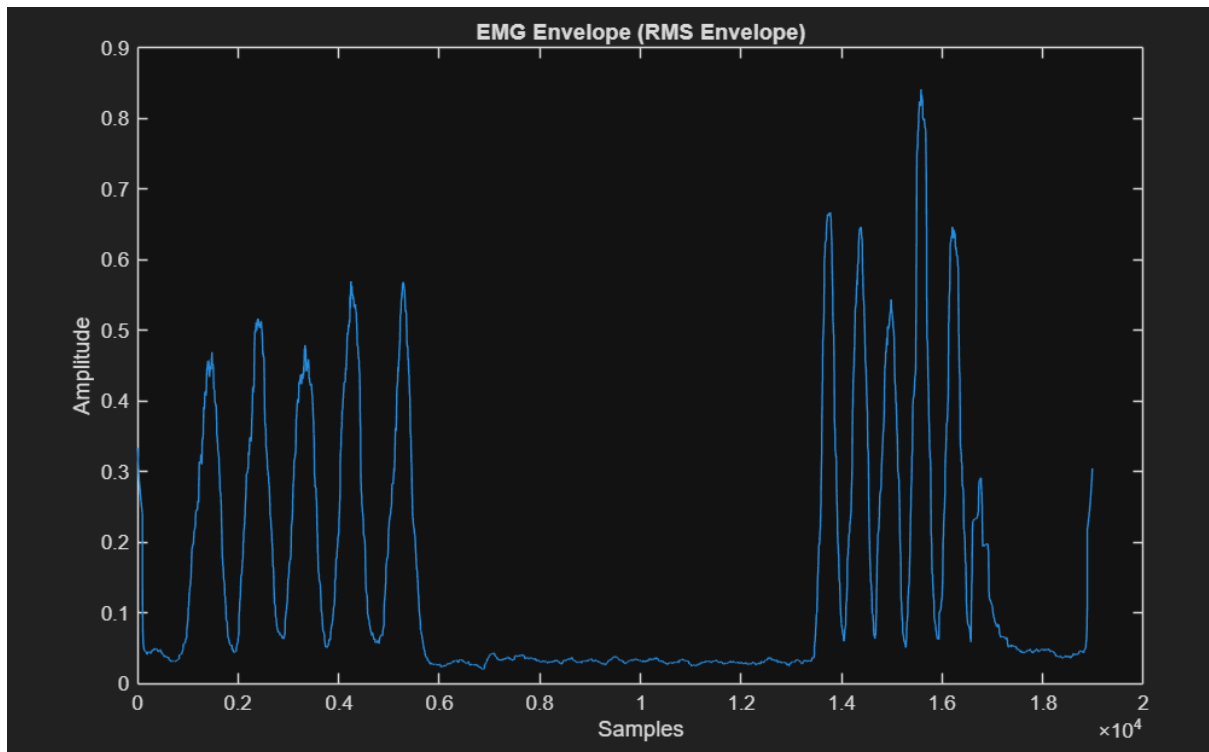


Figure 14: EMG Envelope

Result Discussion

EMG measures muscle electrical activity to quantify activation intensity but does not show direction of muscle action (concentric vs eccentric), unlike IMU gyroscope signals that capture rotational direction through positive-negative waveform patterns.

Figure 14 shows the processed EMG envelope from the vastus lateralis muscle, displaying five distinct activation peaks during slow squats followed by five higher and narrower peaks during fast squats. The signal pattern exhibits periodic bursts synchronized with squat cycles because muscle electrical activity increases precisely when quadriceps contract during the eccentric (downward) and concentric (upward) phases of each repetition, returning to baseline during rest periods between sets. EMG relates directly to movement phases by showing higher amplitude and sustained activation during slow squats (longer muscle effort for control) versus sharper, more intense peaks during fast squats (explosive power generation), confirming that vastus lateralis activation scales with both movement speed and duration as expected in squat biomechanics.

Limitations

- Loose IMU bandage attachment on thigh during first trial introduced motion artifacts and noise
- Sensor placement variations (especially hip's unstable pelvis location) reduced signal clarity compared to thigh
- Single-subject data prevented statistical reliability or inter-individual comparisons
- EMG provided activation intensity but not muscle contraction direction (concentric vs eccentric)
- Manual segmentation between jumps remained subjective despite reference markers
- Soft tissue movement relative to bone affected signal accuracy
- Potential electrode displacement during dynamic squats
- Moderate rather than extreme speed differences limited biomechanical contrast between slow and fast conditions